



FACULTY EVALUATION SYSTEM

In Partial Fulfillment
of the Requirements for the Course of
Application Development and Emerging Technologies

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CHAPTER I

INTRODUCTION

Background of the Study

Faculty evaluation is essential for maintaining high educational standards, as it allows institutions to assess teaching effectiveness and make informed, data-driven decisions. Currently, the institution utilizes basic digital forms to collect feedback; however, this fragmented approach often lacks the integration needed for deep analysis. This results in a process that can feel disconnected for students and time-consuming for administrators who must manually consolidate and interpret the data.

To address these limitations, this study proposes a dedicated Web-Based Faculty Evaluation System designed to make the evaluation process more interactive, engaging, and significantly faster. By moving beyond simple forms to a specialized platform, students can provide feedback through a more intuitive interface, while administrators gain access to instant, automated reporting. Developed using the Waterfall Model, this project follows a disciplined, step-by-step path from initial requirements to final deployment, ensuring a highly reliable and well-documented system that enhances the overall quality of academic assessment.

Project Context

Objectives of the Study

To design and develop a comprehensive web-based Faculty Evaluation and Performance Monitoring System using the Waterfall Model that aims to enhance the efficiency, accuracy, and reliability of the faculty evaluation process by automating data collection, processing, and report generation, while also providing a centralized platform for students to submit evaluations and for administrators to analyze, manage, and utilize evaluation data effectively for informed decision-making and continuous improvement of teaching performance.



This study aims to design, develop, and implement a secure, privacy-preserving online faculty evaluation system for the CIT department of the University of the Assumption. Specifically, it seeks to:

- I. Design and develop a web-based faculty evaluation platform that supports online form submission, data storage, and real-time analytics.
- II. Implement client-side encryption using ECDH (P-256) and AES-GCM (256-bit) to protect the confidentiality of qualitative student feedback before transmission.
- III. Develop an Admin Control Center for managing faculty accounts, evaluation criteria, and evaluation sessions.
- IV. Provide faculty members with a personal dashboard displaying per-criterion analytics and anonymized student feedback.
- V. Restrict system access to verified institutional members through Google OAuth 2.0 domain filtering to ensure only authorized users can participate.

Significance of the Study (Needs revision it is in bullet form based on beneficiaries)

The significance of the study for the faculty evaluation system is summarized below based on the identified beneficiaries and their specific benefits:

- **Students (Users):** They benefit from an engaging interface that allows for quick and convenient online feedback submission from anywhere with an internet connection.
- **Administrators (Users):** They gain access to immediate, automated reporting and real-time monitoring of performance data, which eliminates the need for manual data consolidation and reduces human error.
- **Developers:** They benefit from a structured and reliable development process by using the Waterfall Model and a secure three-tier architecture, ensuring the system is well-documented and scalable.
- **Organization (University of the Assumption):** The institution improves its teaching standards through data-driven decisions while moving toward a more efficient and sustainable digital platform that reduces physical waste and saves time.



Scope and Delimitations

This study focuses on developing a web-based Faculty Evaluation and Performance Monitoring System that allows students to submit evaluations and administrators to manage records and generate reports. The system automates data processing, provides a centralized platform for evaluation, and follows the structured phases of the Waterfall Model, ensuring proper planning, design, implementation, testing, and deployment.

The proposed system is primarily limited by its dependency on a stable internet connection, meaning users in areas with poor connectivity may face difficulties in completing evaluations. Furthermore, while the system streamlines the feedback process, it may not be fully optimized for older hardware or outdated web browsers, which could affect the user experience for some stakeholders. The scope of the platform is also restricted to the internal members of the institution, meaning it is not designed to incorporate external feedback or integrate with third-party academic platforms outside of the current school ecosystem.

CHAPTER II

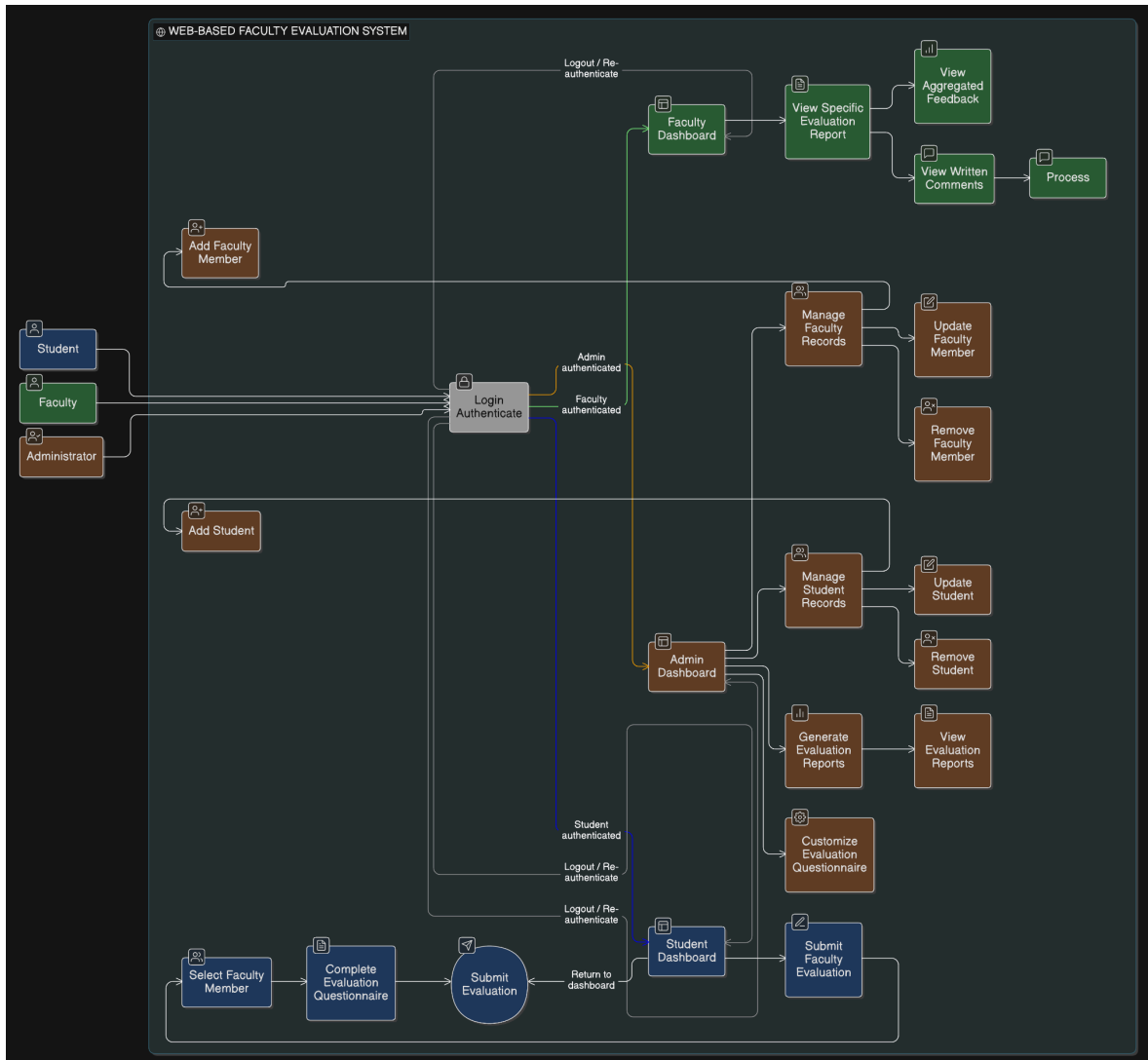
SYSTEM ANALYSIS AND DESIGN

System Overview

The proposed system is a centralized web-based platform designed to automate the faculty evaluation process and enhance the management of evaluation data. It allows students to conveniently submit evaluations online, eliminating the inefficiencies associated with manual or paper-based systems. Administrators can access, monitor, and analyze evaluation results through an integrated reporting feature, which facilitates informed decision-making regarding faculty performance. The system follows the Waterfall Model, ensuring that each development phase from requirements gathering and system design to implementation, testing, and deployment is conducted systematically, thoroughly documented, and executed in a sequential manner to maintain consistency and reliability.

Use Case Diagram

The Web-Based Faculty Evaluation System utilizes a Three-Tier Architecture to ensure a secure and efficient workflow. It consists of a Presentation Layer (React Native) for the user interface, an Application Layer (Java) to handle business logic and score calculations, and a Data Layer (PostgreSQL) for secure record storage. This modular approach enhances security by isolating the database from direct user access and provides scalability, allowing the system to handle increasing amounts of evaluation data without compromising performance.





Use Case Description

Table 1. Use Case Descriptions

Use Case	Actor	Preconditions	Postconditions
Student Login	Student	The student has a valid @ua.edu.ph Google account; the system is online.	Students are authenticated and directed to the Student Identification Page.
Submit Evaluation	Student	The student is logged in and has been identified with a valid student number and section.	Evaluation is encrypted via ECDH and AES-GCM and securely stored in the database.
View Evaluation Results	Faculty	Faculty members are authenticated via @ua.edu.ph Google account.	Faculty can view their performance scores, per-criterion analytics, and anonymized student feedback.
Manage Faculty Accounts	Administrator	Administrator is authenticated and the Admin Control Center is accessible.	Faculty accounts, assigned sections, and roles are created, updated, or removed.
Configure Evaluation Criteria	Administrator	Administrator is logged in to the Admin Control Center.	Evaluation criteria are added, updated, or deleted for use in student evaluation forms.
View Encrypted Feedback	Administrator	Administrator is authenticated and encrypted submissions exist in the database.	Decrypted student feedback is displayed in the Admin Control Center evaluations table.
Manage Evaluation Session	Administrator	Administrator is logged in; faculty and criteria are configured.	Evaluation sessions are activated or deactivated, controlling student access to evaluation forms.

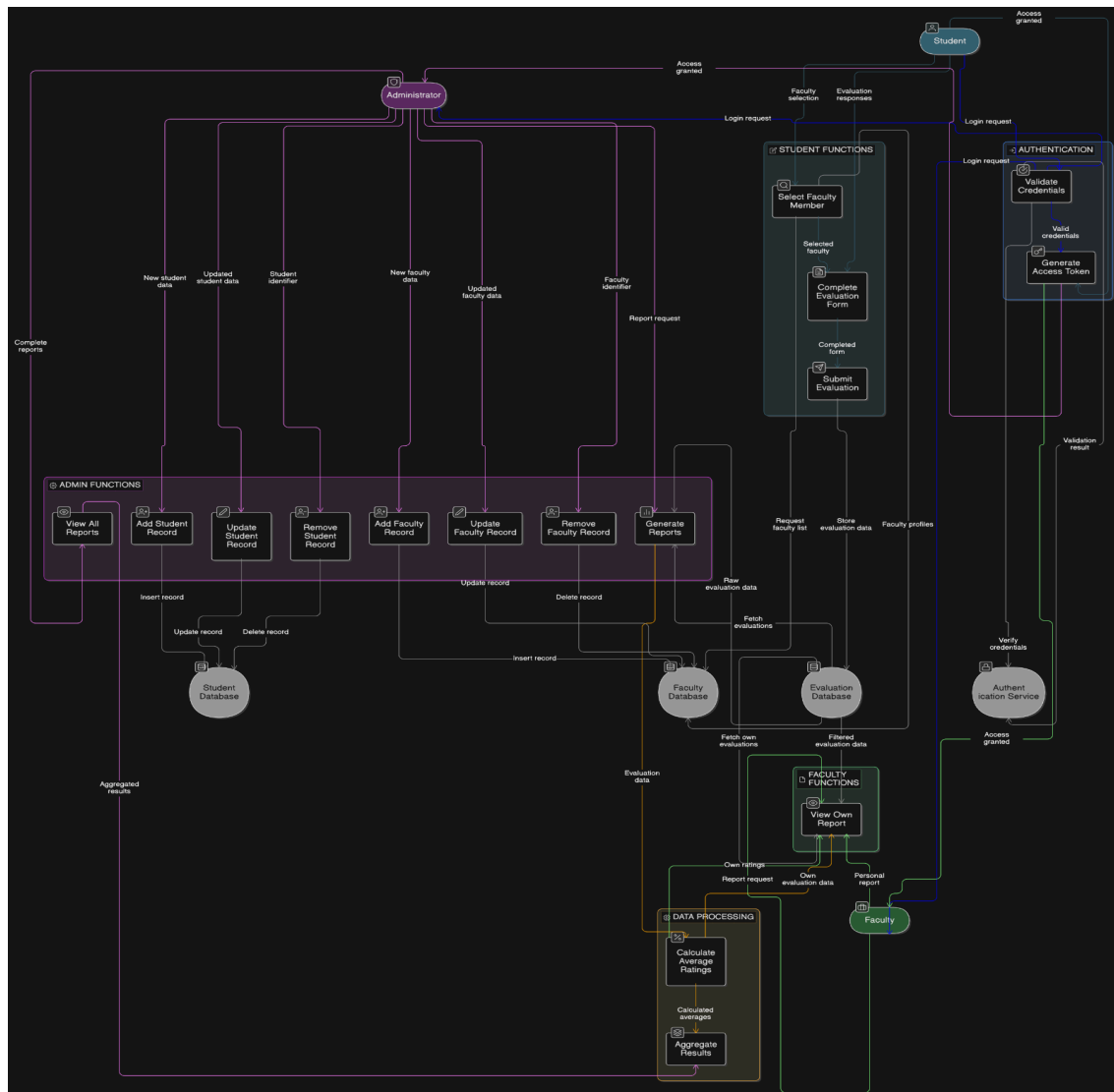
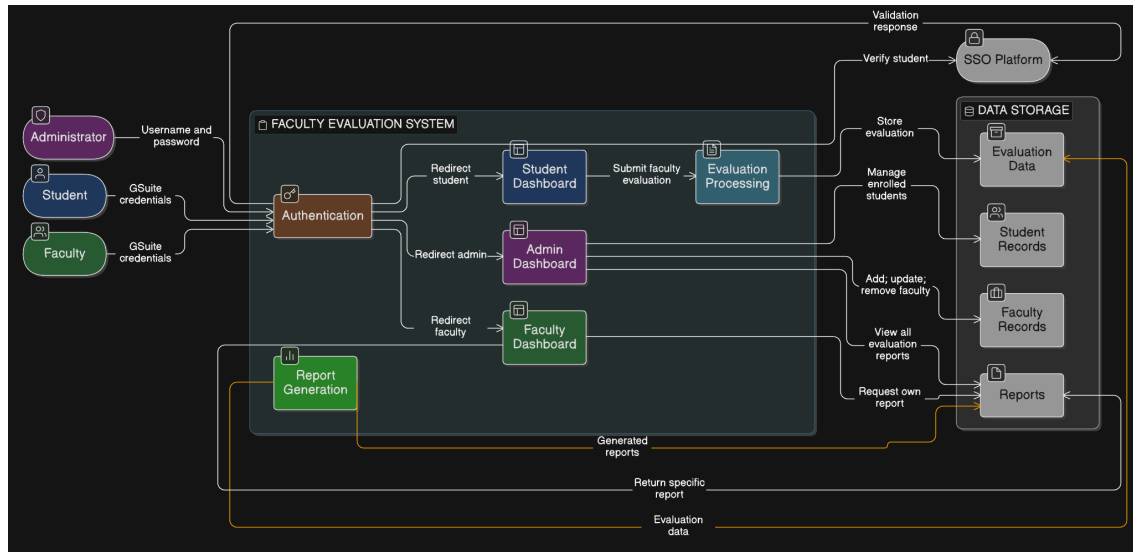
Acceptance Criteria

Table 2. Acceptance Criteria per Use Case

Use Case	Acceptance Criteria
Student Login	Only @ua.edu.ph Google accounts are accepted. Non-institutional accounts are blocked. Successful login redirects the student to the Student Identification Page.
Submit Evaluation	Students must enter a valid student number and select a year/section. Evaluation ratings must be completed for all assigned faculty. Written comments are encrypted in the browser before submission. The system prevents duplicate submissions for the same faculty.
View Evaluation Results	Faculty can view their overall performance score and breakdown per evaluation criterion. Student feedback is displayed anonymously with no identifying information visible.
Manage Faculty Accounts	Administrators can add, edit, and delete faculty accounts. Each faculty member can be assigned to one or more sections. Changes are saved and reflected immediately in the system.
Configure Evaluation Criteria	Administrators can add, rename, and remove evaluation criteria. Changes are applied to all subsequent evaluation forms.
View Encrypted Feedback	Feedback is stored as encrypted ciphertext. Authorized decryption via the server's private EC key reveals the plaintext comment. Decryption fails gracefully if the key is unavailable.
Manage Evaluation Session	Administrators can activate and deactivate the evaluation session. Students can only access evaluation forms when a session is active.

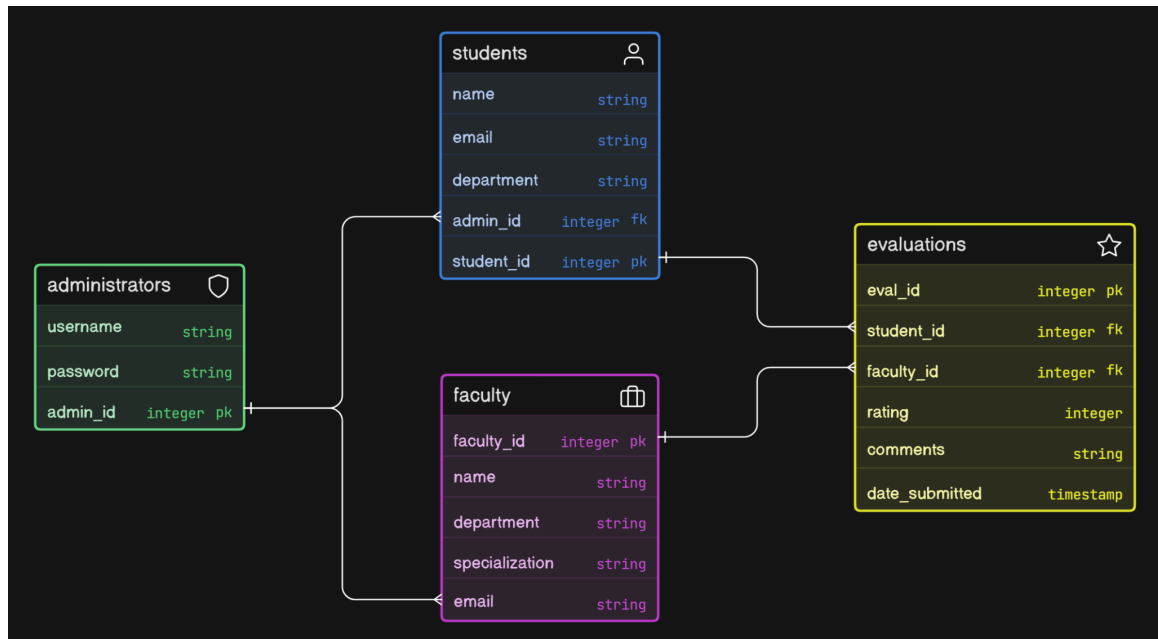
Data Flow Diagram

Insert Context Diagram and Level 1 DFD with explanations.



Entity Relationship Diagram

Insert ERD and explain entities and relationships.



CHAPTER III

METHODOLOGY, PROTOTYPE AND TESTING

Development Methodology

Describe the SDLC/iterative approach used and justify why it fits the project.

Development Process

The development of the Faculty Evaluation System followed the Waterfall Model, ensuring each step was finished before moving to the next. Here is how we executed each phase:

- **Planning:** We started by setting our main goal: replacing the slow manual evaluation process with a better digital version. We defined exactly what the system needed to do to make sure the project stayed on track from start to finish.
- **Analysis:** We gathered all the requirements, focusing on things like secure logins and easy form submissions. We also used data flow diagrams to map out how information would move through the system.
- **Design:** We created a blueprint for the system, choosing a three-tier architecture that separates the user interface, the logic, and the database. This included designing the database structure and how the screens would look.
- **Implementation:** Our team built the actual system using React Native for the front end and Java for the back end. We made sure the "brain" of the system could handle all the data processing and connect smoothly to the PostgreSQL database.
- **Testing:** We used Postman to check the APIs and Selenium IDE to record and replay user actions. By comparing our expected results with how the system actually performed, we were able to find and fix bugs.
- **Evaluation:** In the final phase, we checked if the system truly solved the original problem. We confirmed that the platform is fast, organized, and ready to provide the university with better data for improving teaching quality.

System Architecture

Insert architecture diagram (e.g., Client–Server, MVC) and explain components and data flow.

[Insert Architecture Diagram Here].

Prototype Implementation

The CIT-EVAL prototype consists of three fully functional user portals corresponding to each user role.

Student Evaluation Hub: Students log in via their UA Google account, verify their identity through the Student Identification Page, and rate their assigned faculty across five criteria using a sliding scale. Written comments are encrypted client-side before submission, and a Review Page allows students to confirm all entries before final secure submission.

Admin Control Center: Administrators access a dashboard displaying key metrics such as total submissions, faculty accounts, and active criteria. The portal includes tabs for viewing encrypted evaluation submissions, managing faculty accounts and sections, and configuring evaluation criteria, along with an analytics view showing faculty rankings and section performance trends.

Faculty Portal: Faculty members can view their overall performance score, student participation count, a Criteria Average Radar chart highlighting strengths and areas for improvement, and a Section Trends chart comparing performance across sections. Anonymized student feedback with decrypted comments is also displayed below.



ACTIVE SESSION
A Student Evaluation Hub
abcluu.studentgua.edu.ph

STUDENT

Faculty Evaluation

Your feedback is private, anonymous, and secured by Diffie-Hellman encryption.

1 Student Info

2 Assigned Faculty Evaluations

3 Review

2. Assigned Faculty Evaluations

Faculty for section 3-A are pre-assigned. Use each tab to complete metrics and secure feedback.

Jonilo Mababa Pending

Fernand Layug Pending

Lincoln Vinuya Pending

1 >

Jonilo Mababa

Role: Professor

Complete all metric scores for this faculty tab. Each criterion requires an explicit score selection.

Clarity and Communication

5

ACTIVE SESSION
A Administrator Control Center
adminsystem

ADMIN

Admin Control Center

Manage encrypted evaluations, professors, criteria, and analytics.

Refresh

Export CSV

Export PDF

Preview faculty page

ENCRYPTED FEEDBACK

62

Stored submissions

FACULTY ACCOUNTS

11

Managed professors

EVALUATION CRITERIA

5

Active metrics

Overall

Evaluations

Professors

Criteria

Overall Evaluation Insights

Track section performance and faculty ranking in one responsive view.

Section: All Sections

SUBMISSIONS

62

Across all sections

AVERAGE SCORE

7.26

Across selected scope

RANKED TEACHERS

10

With at least one response

BEST TEACHER

10

Anton Lulu

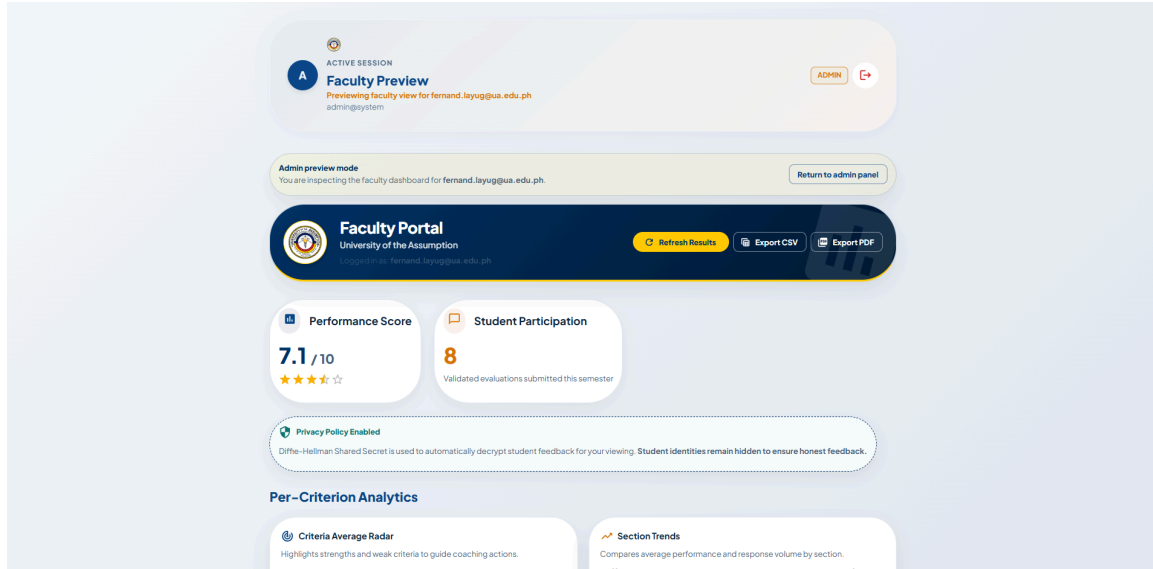
Section Score Trend

Line chart of average teacher scores per section.

Teacher Evaluation Share

Pie chart of response share among top-ranked teachers.

11



System Features (Implemented)

Table 3. Implemented System Features

Feature	Description	Status
Google OAuth 2.0 Login	Domain-restricted authentication allowing only @ua.edu.ph accounts to access the system.	Working
Student Identification	Validates student number and year/section before allowing access to evaluation forms.	Working
Client-Side Encryption	ECDH (P-256) key exchange and AES-GCM (256-bit) encryption of qualitative feedback in the browser before transmission.	Working
Evaluation Form Submission	Students rate assigned faculty across five criteria using a sliding scale and optionally provide written feedback.	Working
Duplicate Submission Prevention	The system tracks whether a student has evaluated a faculty member and blocks resubmission without linking identity to content.	Working



Admin Control Center	Dashboard for managing faculty accounts, sections, criteria, evaluation sessions, and viewing submissions.	Working
Faculty Performance Dashboard	Personal portal showing overall score, criteria radar chart, section trends, and anonymized student feedback.	Working
Evaluation Criteria Management	Administrators can add, edit, and delete evaluation criteria used in student forms.	Working
Encrypted Feedback Decryption	Server-side decryption using the private EC key to reveal student comments for authorized faculty.	Working
Real-Time Analytics	Admin dashboard displays faculty rankings, section performance trends, and evaluation completion statistics.	Working

Testing Methodology

To conduct User Acceptance Testing (UAT), the team first identified the core functions of the system, such as Student Login, Evaluation Submission, and Admin Faculty Management, and defined them as Use Cases. From these, the Acceptance Criteria, the specific conditions the system must satisfy to be considered functioning correctly, were used to create detailed test cases that checked for both successful outcomes and common error scenarios. Selenium IDE was used to record user interactions in the web browser and replay them to verify whether the system responded as expected. The Expected Output for each test was then compared against the actual system behavior to identify bugs, confirm correct functionality, and ensure the platform was ready for end users.

Test Case



Test ID	Use Case	Input	Expected Output
TC-01	Student Login	Valid @ua.edu.ph Google account credentials.	Students are authenticated and redirected to the Student Identification Page.
TC-02	Student Login	Non-@ua.edu.ph Google account credentials.	Access is denied. The system displays an unauthorized domain error message.
TC-03	Student Identification	Valid student number and year/section (e.g., 2023000M01, 3-A).	Students proceed to the Evaluation Form for their assigned faculty.
TC-04	Student Identification	Invalid or unrecognized student number.	The system displays a validation error and prevents progression.
TC-05	Submit Evaluation	All criteria rated; qualitative comment entered and submitted.	Comment is encrypted client-side via ECDH/AES-GCM; encrypted record is stored in the database.
TC-06	Submit Evaluation	Student attempts to re-evaluate an already-evaluated faculty.	The system detects duplicate submission and blocks resubmission with an appropriate message.
TC-07	View Faculty Dashboard	Faculty members log in with a valid @ua.edu.ph account.	Faculty portal loads displaying performance score, criteria radar chart, and anonymized feedback.
TC-08	Admin — Add Faculty	The administrator enters a new faculty name, email, role, and assigned sections.	A new faculty account is created and appears in the Professors list.
TC-09	Admin — Delete Faculty	Administrator clicks Delete on an existing faculty entry.	The faculty account is removed from the system and no longer appears in the list.
TC-10	Admin — View Feedback	Administrator accesses the Evaluations tab.	Feedback is decrypted server-side and displayed in the evaluations table alongside performance scores.

Test Results

Table 5. Test Results

Test ID	Expected	Actual	Status
TC-01	Student authenticated and redirected.	Student authenticated and redirected.	Passed
TC-02	Access denied for non-UA accounts.	Access denied for non-UA accounts.	Passed
TC-03	The student proceeds to the evaluation form.	The student proceeds to the evaluation form.	Passed
TC-04	Validation error displayed.	Validation error displayed.	Passed
TC-05	Feedback encrypted and stored.	Feedback is encrypted and stored as ciphertext.	Passed
TC-06	Resubmission blocked.	Resubmission blocked with message.	Passed
TC-07	The faculty dashboard loads correctly.	Faculty dashboard loads with all analytics.	Passed
TC-08	New faculty created successfully.	New faculty created and visible in the list.	Passed
TC-09	Faculty removed from list.	Faculty removed from list.	Passed
TC-10	Decrypted feedback displayed.	Feedback displayed in the evaluations table.	Passed

Issues and Improvement

Table 6. Identified Issues and Proposed Fixes

Issue / Bug	Description	Proposed Fix
Browser Compatibility	The Web Crypto API (used for ECDH and AES-GCM encryption) is not supported on older or outdated browsers, causing the encryption process to fail silently.	Add a browser compatibility check on page load and display a clear warning message to users on unsupported browsers, prompting them to update or switch to a supported browser.
Private Key Dependency	If the server's private EC key is lost or corrupted, all historically encrypted feedback becomes permanently unrecoverable with no fallback mechanism.	Implement a secure key backup policy using a Hardware Security Module (HSM) or encrypted key escrow solution to allow key recovery in authorized scenarios.



No Response Validation	The system cannot verify whether students are providing sincere or accurate responses, as it only tracks submission status, not response quality.	In future iterations, consider implementing response quality heuristics or minimum character requirements for written feedback to improve data usefulness.
Limited to CIT Department	The current deployment is restricted to the CIT department and does not support the evaluation structures or criteria of other colleges.	Design a multi-department configuration module that allows administrators from different colleges to manage their own evaluation sessions independently within the same platform.
Session Management	There is no automated mechanism to open or close evaluation sessions based on a schedule; administrators must manually activate and deactivate sessions.	Implement a scheduled session management feature that allows administrators to set start and end dates and times for evaluation periods, with automated activation and deactivation.

Tools and Technologies

Table 7. Tools and Technologies Used

Component	Technology / Tool	Purpose
Backend Framework	Java 17, Spring Boot 3.4.4	Handles REST API endpoints, business logic, session management, and server-side ECDH decryption.
Frontend Library	React 19, Vite, TypeScript	Builds the Single Page Application (SPA) and executes client-side cryptographic operations via the Web Crypto API.
Primary Database	PostgreSQL (with citext)	Stores user accounts, evaluation records, encrypted feedback blobs, faculty data, and evaluation criteria.
UI Framework	Material UI (MUI), Framer Motion	Provides styled, accessible UI components and smooth animations for the student, faculty, and admin portals.



Authentication	Google OAuth 2.0, JWT	Restricts system access to @ua.edu.ph domain accounts and manages session tokens.
Encryption Standard	Web Crypto API: ECDH (P-256), AES-GCM (256-bit)	Performs client-side key generation, shared secret derivation, and authenticated encryption of qualitative student feedback.
Testing Tool	Selenium IDE	Records and replays user interactions in the browser to automate UAT test case execution.
Version Control	GitHub	Hosts the frontend and backend source code repositories for collaborative development and version tracking.
Build Tool	Maven (Backend), npm / Vite (Frontend)	Manages dependencies and builds the backend JAR and frontend SPA for deployment.

Prototype Results and Future Enhancements

The CIT-EVAL prototype is a secure, three-tier web application built on React Native, Java, and PostgreSQL that has already cleared its core User Acceptance Testing. It streamlines faculty evaluations by restricting access to verified university accounts via Google OAuth 2.0 and generating real-time performance metrics while strictly maintaining participant anonymity. To ensure total confidentiality, the system employs advanced client-side encryption, specifically ECDH (P-256) and AES-GCM (256-bit) to secure qualitative feedback before it is ever transmitted to the server.

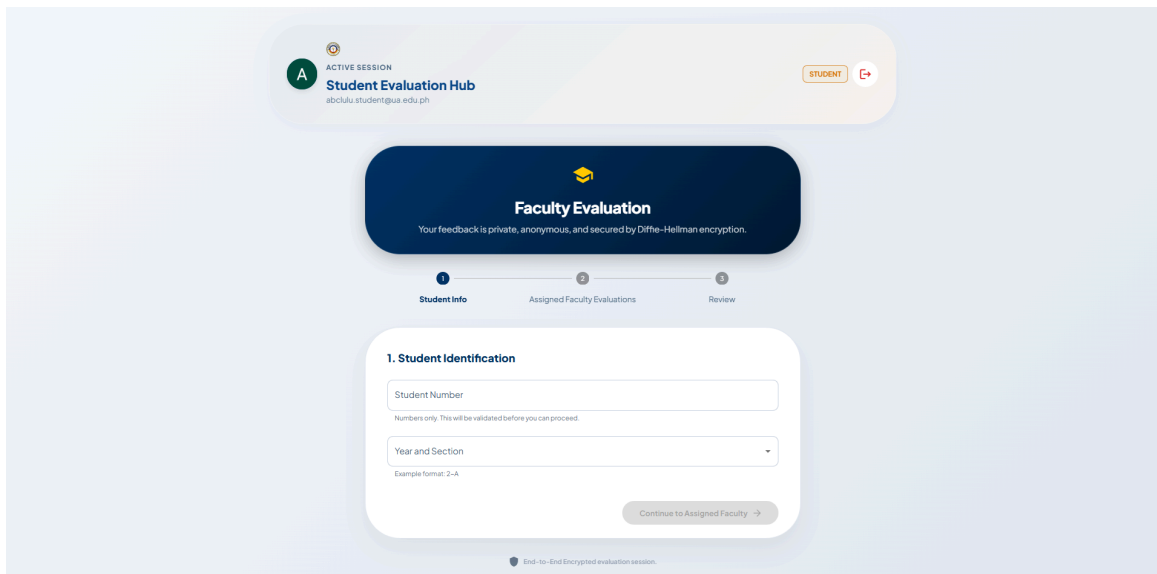
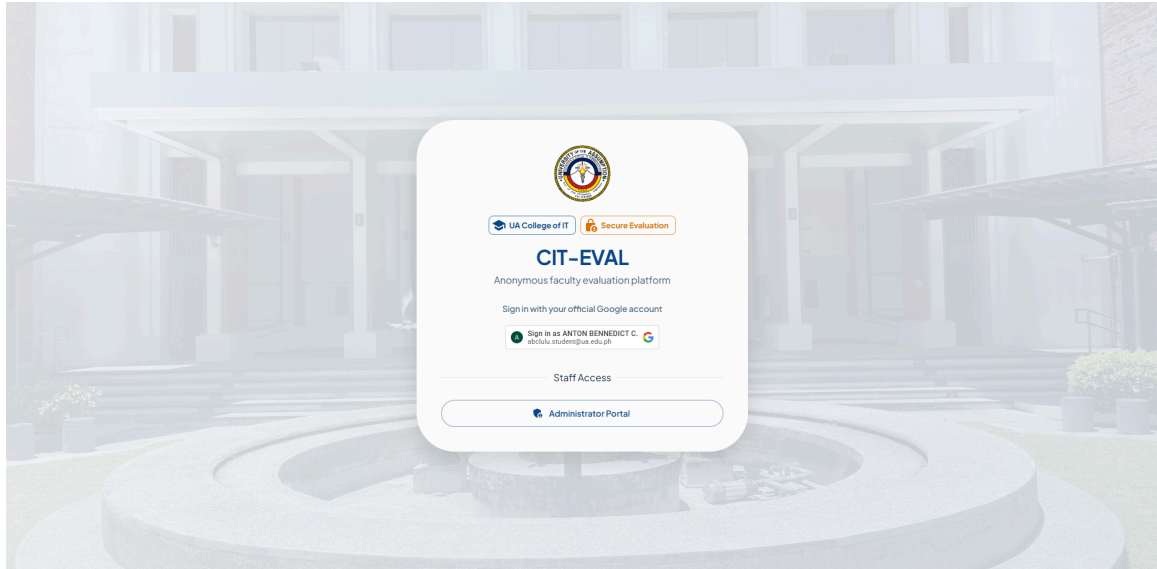
Designed for long-term scalability, the platform utilizes a decoupled architecture and standard repository controls to manage high traffic volumes during peak evaluation windows. Moving forward, the project aims to transition into a multi-department module that allows different colleges within the University of the Assumption to run independent evaluation cycles simultaneously. Planned upgrades include automated session scheduling to reduce manual administration, text-validation heuristics to improve the quality of feedback, and interface optimizations to support users on older hardware or unstable internet connections.



APPENDIX A: Additional Test Cases

APPENDIX B: Screenshots

Insert additional screenshots (UI, testing, errors, etc.).





ACTIVE SESSION
A
Student Evaluation Hub
abcluu.studentgua.edu.ph

STUDENT

Faculty Evaluation

Your feedback is private, anonymous, and secured by Diffie-Hellman encryption.

1

2

3

Student InfoAssigned Faculty EvaluationsReview

2. Assigned Faculty Evaluations

Faculty for section 3-A are pre-assigned. Use each tab to complete metrics and secure feedback.

Jonilo Mababa PendingFernand Layug PendingLincoln Vinuya Pending1 >

Jonilo Mababa

Role: Professor

Complete all metric scores for this faculty tab. Each criterion requires an explicit score selection.

Clarity and Communication5

ACTIVE SESSION
A
Administrator Control Center
adminsystem

ADMIN

Admin Control Center

Manage encrypted evaluations, professors, criteria, and analytics.

RefreshExport CSVExport PDFPreview faculty page

ENCRYPTED FEEDBACK
62
Stored submissions

FACULTY ACCOUNTS
11
Managed professors

EVALUATION CRITERIA
5
Active metrics

OverallEvaluationsProfessorsCriteria

Overall Evaluation Insights

Track section performance and faculty ranking in one responsive view.

SectionAll Sections

SUBMISSIONS
62
Across all sections

AVERAGE SCORE
7.26
Across selected scope

RANKED TEACHERS
10
With at least one response

BEST TEACHER
10
Anton Lulu

Section Score Trend

Line chart of average teacher scores per section.

Teacher Evaluation Share

Pie chart of response share among top-ranked teachers.



admin@system

Admin Control Center

Manage encrypted evaluations, professors, criteria, and analytics.

[Refresh](#) [Export CSV](#) [Export PDF](#) [Preview faculty page](#)

ENCRYPTED FEEDBACK

62

Stored submissions

FACULTY ACCOUNTS

11

Managed professors

EVALUATION CRITERIA

5

Active metrics

Overall **Evaluations** Professors Criteria

Student Email	Student ID	Faculty Email	Performance	Feedback	Security
bongalang.student@...	2023001103	lincoln.vinuyagua.edu.ph	5.0	tnWCS79Pcx66sTechhLR+9VVCkRcbyPttlaD8kJOHv	Decrypt
bongalang.student@...	2023001103	maribeth.rivera@ua.edu...	5.0	LXZBshTmTb/HfLuxUmH2LDZDUiewMpiEowzO9ehCaeBNO	Decrypt
bongalang.student@...	2023001103	marstephen.sembrano@...	5.0	xHiaCvLVExy/lq73yulW+fgQpwZD+yRTTWVWsdvashHnq	Decrypt
anlaxamana.student@...	12312312	Antongua.edu.ph	10.0	Vw+CEaHSGuPvash4H2HwY7mMLZ7wMDaHANI24N9RwagA9Q	Decrypt
anlaxamana.student@...	12312312	grace.tiquig@ua.edu.ph	7.6	+d7F6esBRpu6aomjyLMfnkwp7JlQ+=	Decrypt
gepastoral.student@...	2021000022	jonilo.mababaga@ua.edu...	10.0	cLxFT6exIT9/cOSv4MZOHCITDhamDddb8BzSlug16Id7AP	Decrypt
gepastoral.student@...	2021000022	mark.madalipay@ua.edu...	10.0	VOk+FFFDIZYVFFQz+xZOUd+Awk89mU8rbzD96VnqJlI	Decrypt
ajdalanzo.student@ua...	2023000905	marstephen.sembrano@...	8.0	Uua9H13W65+Iqu57/7Z7q10B83/t9dRkuOmRQZQZQz+	Decrypt
ajdalanzo.student@ua...	2023000905	innin.mahabaga@ua.edu...	8.0	...	Decrypt

Rows per page: 10 1-10 of 62

ACTIVE SESSION

Faculty Preview

Previewing faculty view for fernand.layug@ua.edu.ph

admin@system

[ADMIN](#) [+](#)

Admin preview mode

You are inspecting the faculty dashboard for fernand.layug@ua.edu.ph

[Return to admin panel](#)

Faculty Portal

University of the Assumption

[Refresh Results](#) [Export CSV](#) [Export PDF](#)

Performance Score

7.1 / 10

★★★★☆

Student Participation

8

Validated evaluations submitted this semester

Privacy Policy Enabled

Diffie-Hellman Shared Secret is used to automatically decrypt student feedback for your viewing. Student identities remain hidden to ensure honest feedback.

Per-Criterion Analytics

Criteria Average Radar

Highlights strengths and weak criteria to guide coaching actions.

Section Trends

Compares average performance and response volume by section.